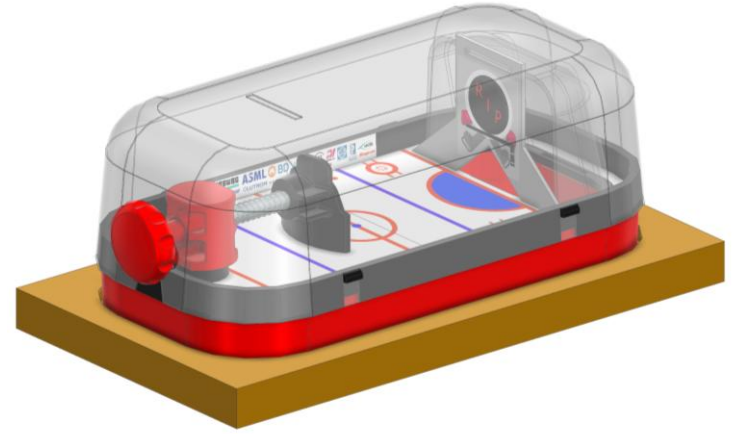


RPI



Handheld Hockey

MANE 4610 Team D Fall 2025

Team Members



Alex Wallach

Robotics Engineer/Part
Transfer Engineer



Cole Delage

Processes
Engineer



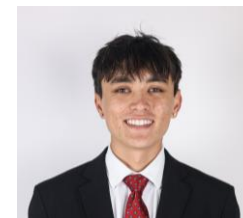
Julia Tsao

Financial
Officer/Robotics
Engineer



Megan Lin

Assembly
Manager/Hard
Automation Engineer



Julian Fong-McAdams

Processes Engineer/
Packaging
Engineer



Emma Crane

Plastics Engineer/Hard
Automation Engineer



Paul Sussman

Plastics Engineer/
Packaging Engineer



Muthashim Alam

Manufacturing
Manager



Joey Marchand

Information Manager



Elliot So

Project Manager/Hard
Automation Engineer

Agenda

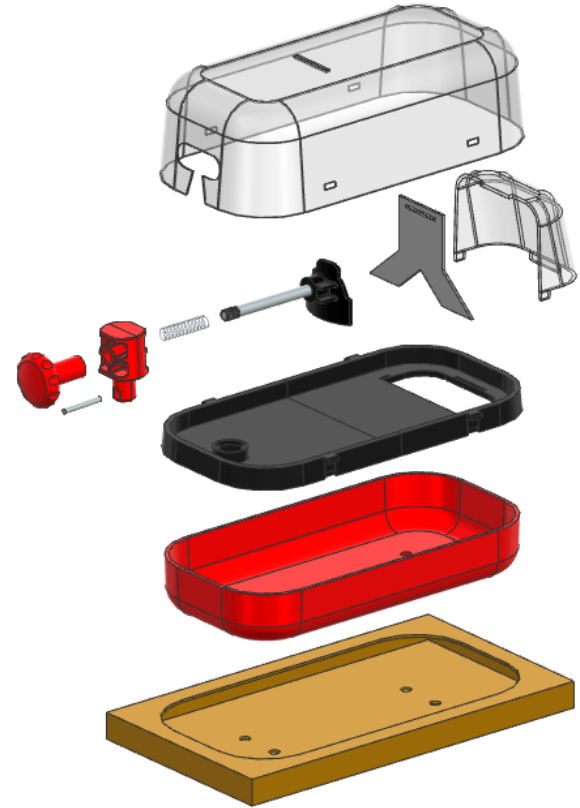
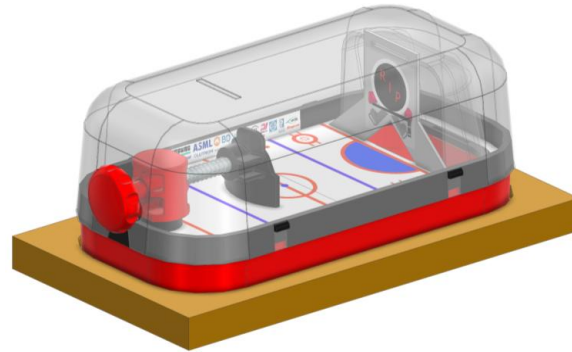
1. Team Introduction
2. Project Overview
3. Major Design Changes from Mid-Term
4. Prototype Demonstration
5. Manufacturing Process
6. Assembly Process
7. Manufacturing & Assembly Challenges
8. Budget Forecast
9. Proposed Project Plan for Spring Semester
10. Lessons Learned

Project Overview

Coin bank in the style of a hockey game
User shoots coins into goal with the launcher

Notable Updates:

- Finalized mechanical design
- Finalized assembly processes



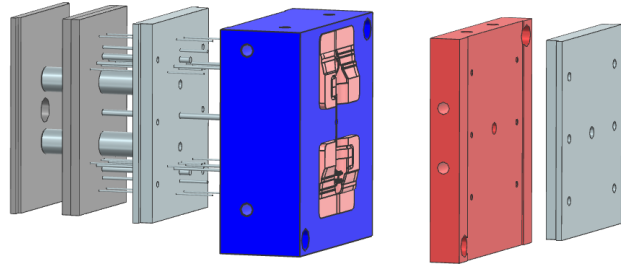
Prototype Demonstration



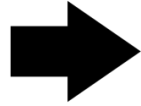
Manufacturing Processes Overview

Manufacturing Process	Parts
Plastic Injection Molding (PIM)	Goal, Turret, Puckman, Lid
Vacuum Forming	Bubble
Laser Cutting	Goalie, Cardboard Packaging
Abrasive Water Jet Cutting	Base
Deep Drawing	Base
CNC Machining	Handle, Bolt, Base

Manufacturing - Plastic Injection Molding (ABS+HDPE)



3x4 PIM assy



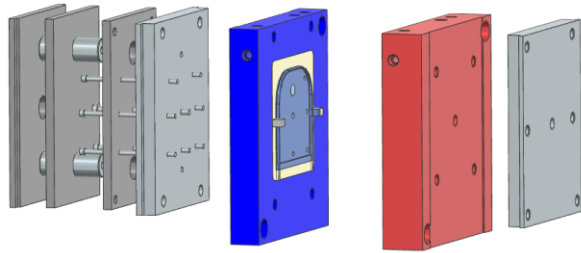
Turret



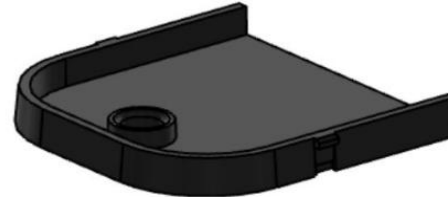
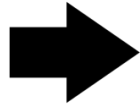
Puckman



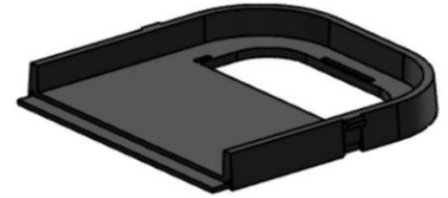
Goal



5x5 PIM assy

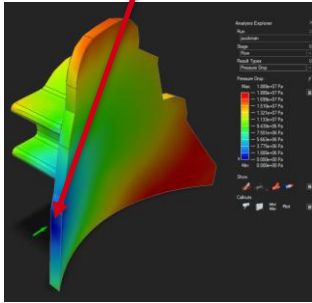
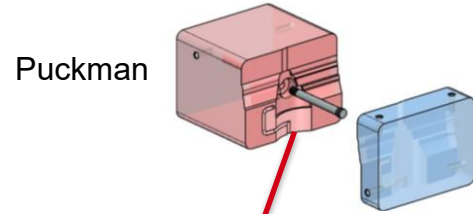


Lid_Launcher

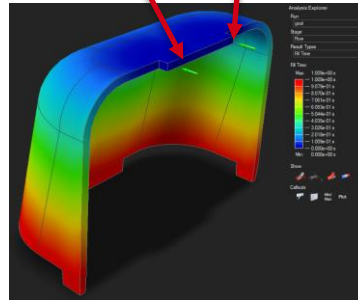
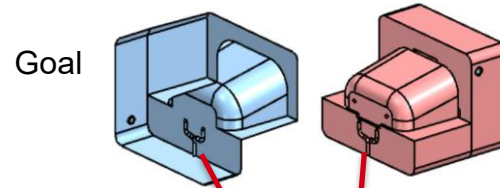


Lid_Goal

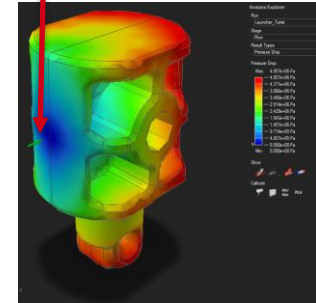
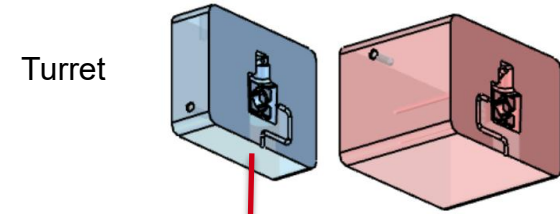
Manufacturing - 3x4 PIM Assembly (Tooling & Simulations)



- Fill Time: 1.02s
- Consistent Wall Thickness
- 0 Weld Lines
- Draft Angles: $\geq 2^\circ$

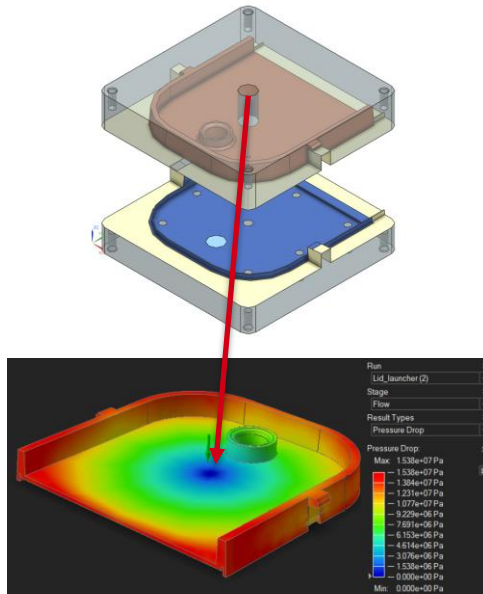


- Fill Time: 1.01s
- Consistent Wall Thickness
- 0 Weld Lines
- Draft Angles: $\geq 2^\circ$

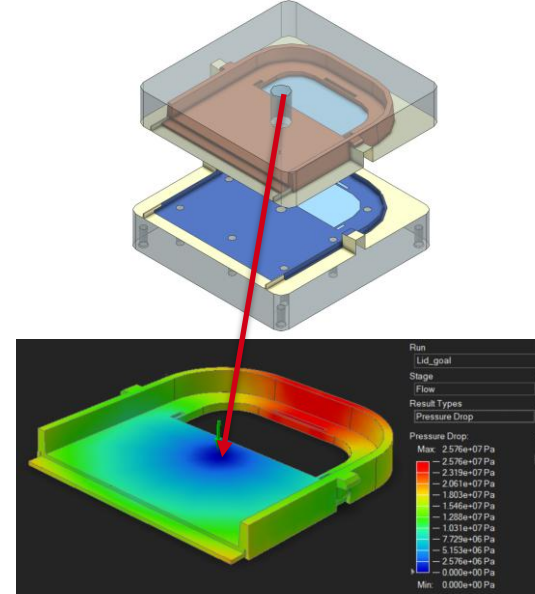


- Fill Time: 0.67s
- Consistent Wall Thickness
- 0 Weld Lines
- Draft Angles: $\geq 2^\circ$

Manufacturing – 5x5 PIM Assembly (Tooling & Simulations)

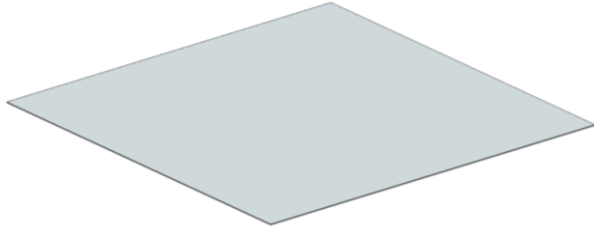


- Fill Time: 1.019s
- Consistent Wall Thickness
- 0 Weld Lines
- Draft Angles: $\geq 2^\circ$

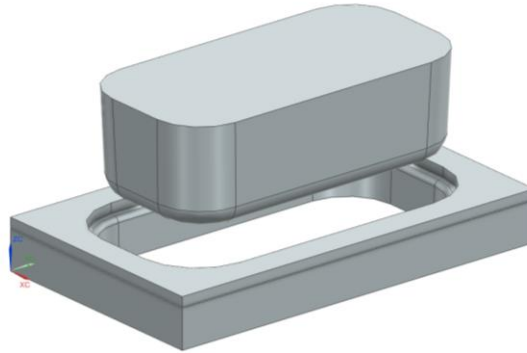


- Fill Time: 1.01
- Consistent Wall Thickness
- 0 Weld Lines
- Draft Angles: $\geq 2^\circ$

Manufacturing Process – Deep Drawing (Base – A1008 Steel)



Blank
12" x 12" 24 Gauge
A1008 Steel

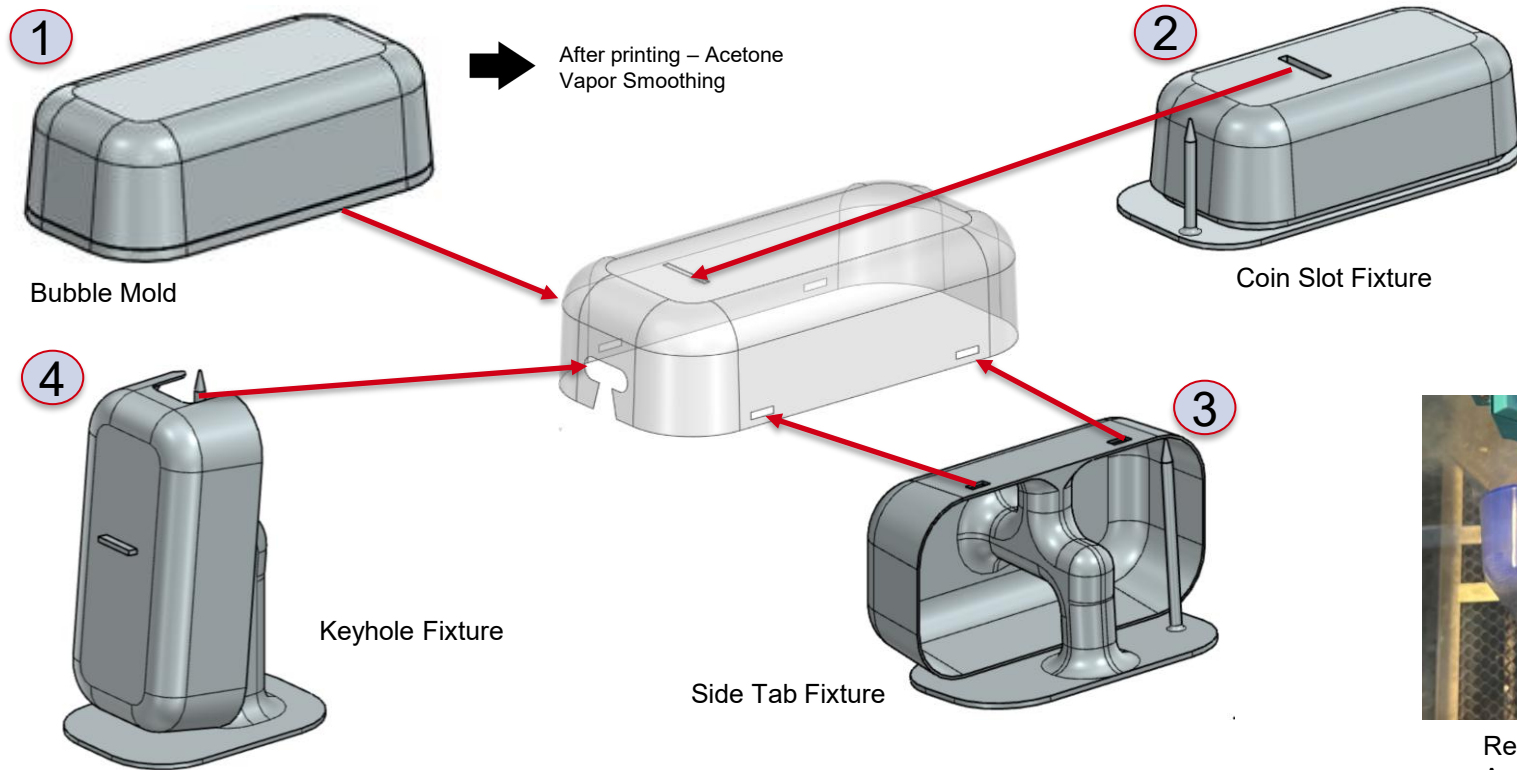


Punch & Die
Deep Drawing



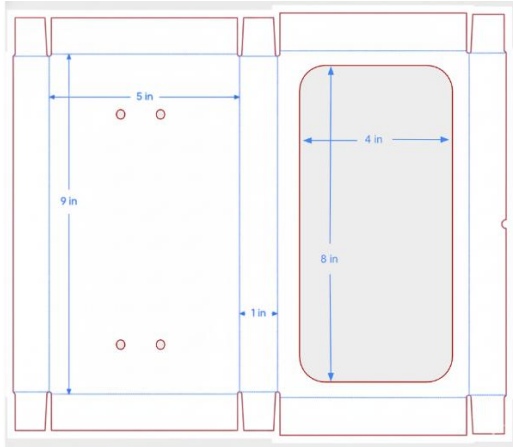
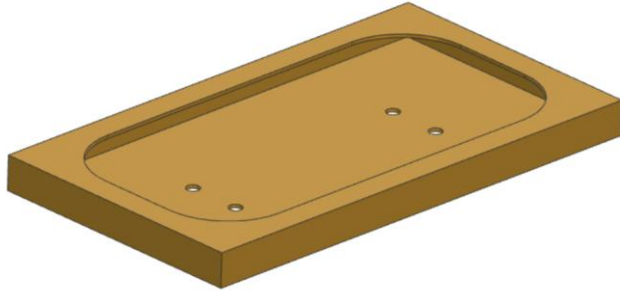
Base
CNC Trimming
Drilling
Powder Coating

Manufacturing – Vacuum Forming & Laser Cut (Bubble - PETG)

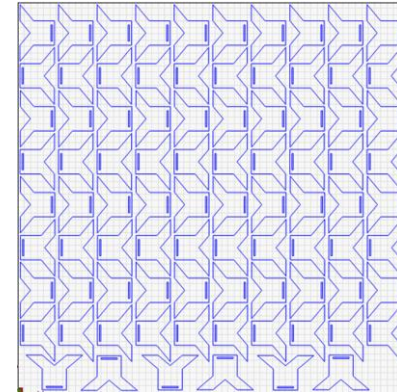


Real World Application

Manufacturing – Laser Cutting (Goalie & Packaging)



Packaging Laser Cutting Layout



24" x 24" Goalie Laser Cutting Layout

Manufacturing Process

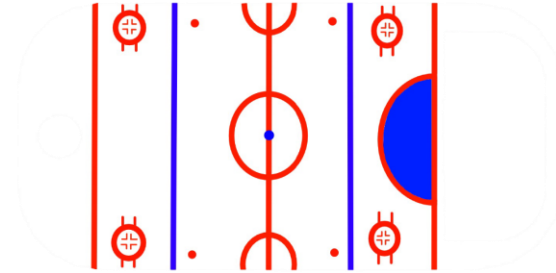
Part Manufacturing:

Part Name: Rink Sticker, Goalie Sticker, Puckman Sticker, Sponsor Sticker

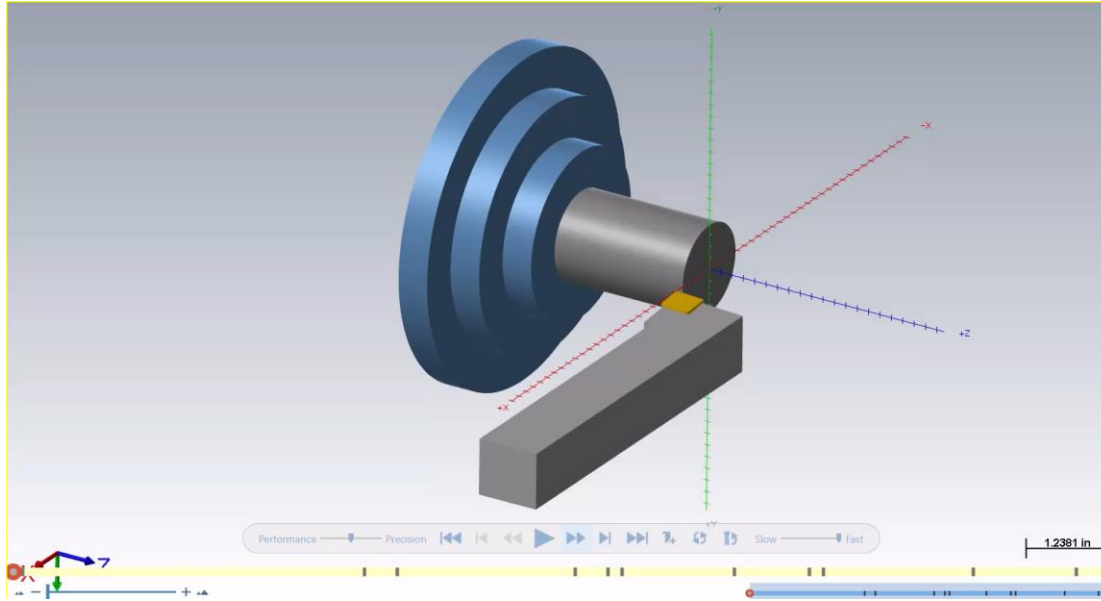
Material: Vinyl

Process:

1. Sticker Printing



Manufacturing Process - CNC Turning (Handle – 6061 Aluminum)



Handle
6061 Aluminum

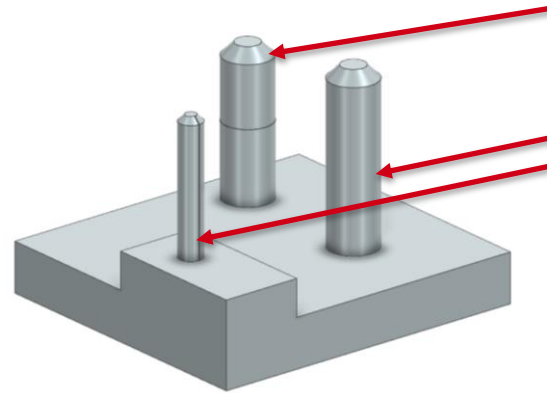
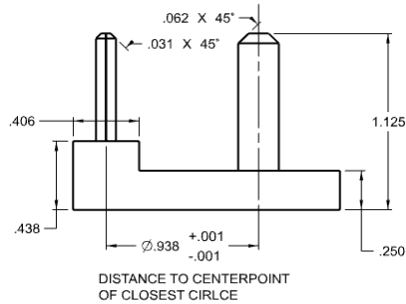
Manufacturing Process – CNC Turning (Bolt – 6061 Aluminum)



Nominal Size or Basic Product Diameter	Full-Size Body Diameter, <i>E</i> (See Paras. 3.4 and 3.5)	
	Max.	Min.
1/4	0.2500	0.260
5/16	0.3125	0.324
3/8	0.3750	0.388

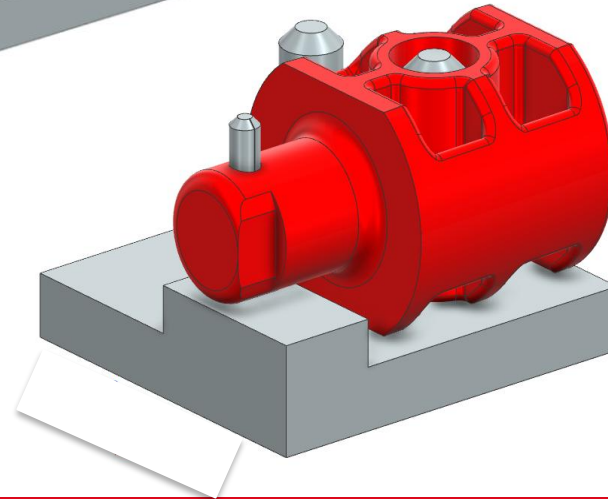
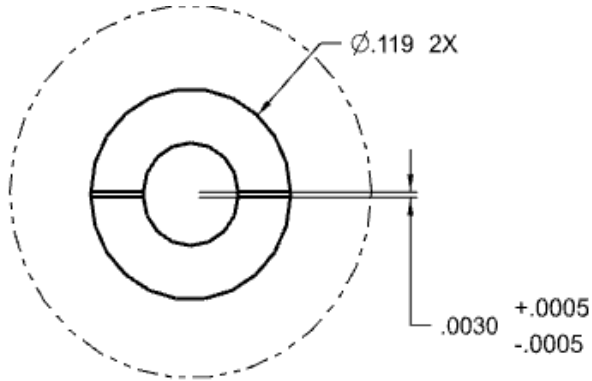
Nominal Size Range, Inches Over To	Class RC1			Class RC2			Class RC3			Class RC4		
	Clear- ance*	Standard Tolerance Limits		Clear- ance*	Standard Tolerance Limits		Clear- ance*	Standard Tolerance Limits		Clear- ance*	Standard Tolerance Limits	
		Hole H5	Shaft g4		Hole H6	Shaft g5		Hole H7	Shaft f6		Hole H8	Shaft f7
Values shown below are in thousandths of an inch.												
0–0.12	0.1 0.45	+0.2 0	–0.1 –0.25	0.1 0.55	+0.25 0	–0.1 –0.3	0.3 0.95	+0.4 0	–0.3 –0.55	0.3 1.3	+0.6 0	–0.3 –0.7
0.12–0.24	0.15 0.5	+0.2 0	–0.15 –0.3	0.15 0.65	+0.3 0	–0.15 –0.35	0.4 1.12	+0.5 0	–0.4 –0.7	0.4 1.6	+0.7 0	–0.4 –0.9
0.24–0.40	0.2 0.6	+0.25 0	–0.2 –0.35	0.2 0.85	+0.4 0	–0.2 –0.45	0.5 1.5	+0.6 0	–0.5 –0.9	0.5 1.0	+0.9 0	–0.5 –1.1

Tolerance Analysis & QC Process Example: Turret

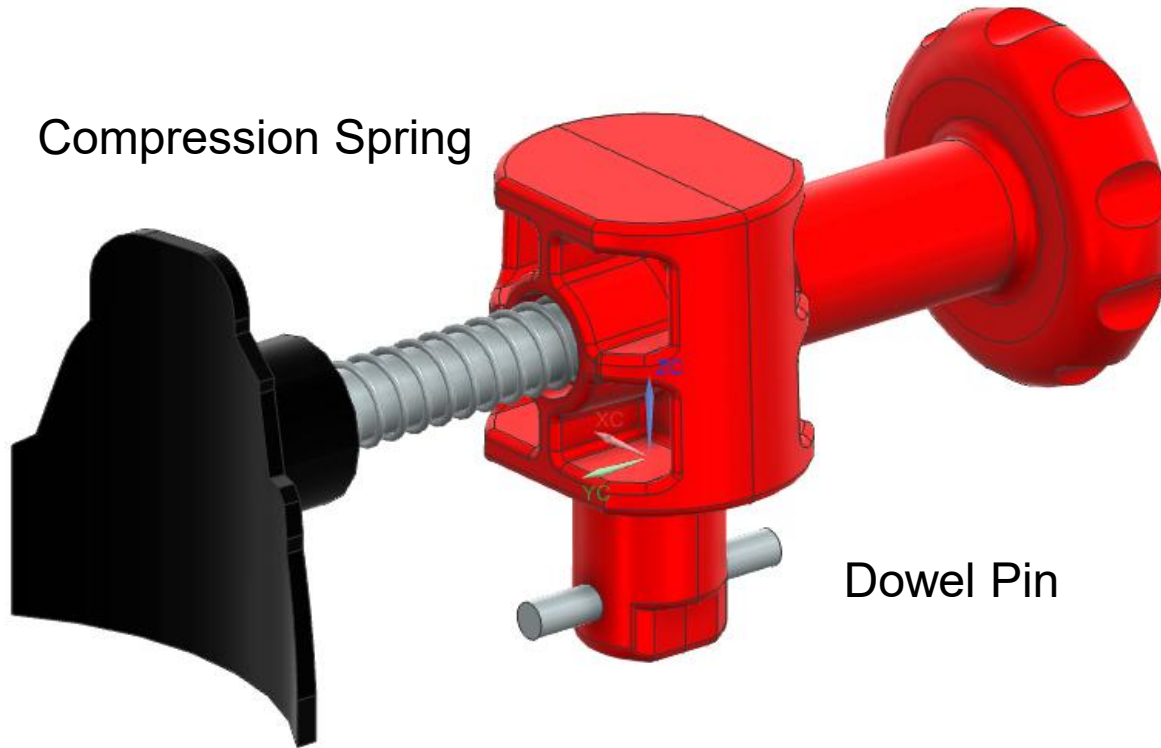


Feature A: Bore Gauge
 ± 0.001 " Diameter

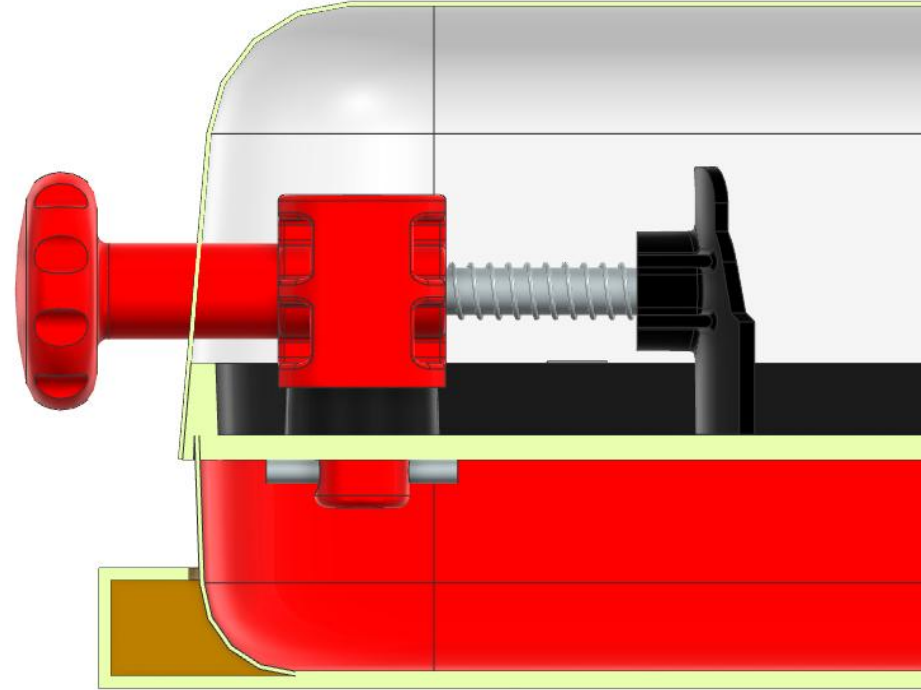
Feature B: Pin Location
Gauge
 ± 0.003 " Distance



Purchased Components



- **Challenge:** Few options for high strength, tight tolerance joining methods that still allow rotation
- **Solution:** A cross pin meets these requirements, but needs specialized assembly mechanisms



Spring Rate Calculations

Maximum desired coin speed: 10 m/s

Mass of dime: 0.0023 kg

Mass of dollar coin: 0.0081 kg

Distance between launcher and goal: 0.127 m

Maximum spring compression: 0.0254 m

Find the friction force acting on coins:

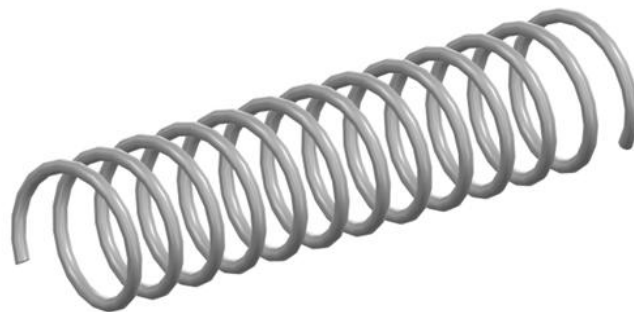
$$\mu = \tan(22^\circ) = 0.40$$

$$f_f = \mu mg = 0.4 \cdot 0.0023 \cdot 9.81 = 0.009 \text{ N}$$

$$\frac{1}{2}kx^2 - f_f d = \frac{1}{2}mv^2$$

Solve for spring rate:

$$k = \frac{mv^2 + 2f_f d}{x^2}$$
$$k = 360 \frac{\text{N}}{\text{m}} = 2.1 \frac{\text{lb}}{\text{in}}$$

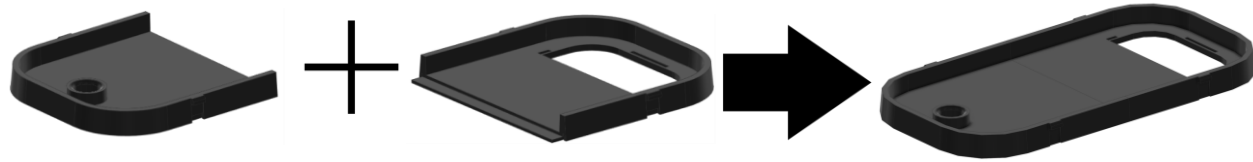


Assembly Process Overview

Assembly Process	Processes
1. Lid Assembly	1. Ultrasonic Welding 2. Manual Sticker Application
2. Goal Assembly	1. Goal to Goalie Automated Press Fit 2. Goal to Lid Automated Press Fit
3. Launcher Assembly	1. Manual Spring Application 2. Automated Turret Attachment 3. Automated Handle Attachment 4. Automated Pin Insertion
4. Base Assembly	1. Manual Twist Tied
5. Coin Bank Assembly	1. Lid to Base Press Fit 2. Manual Sticker + Bubble Application

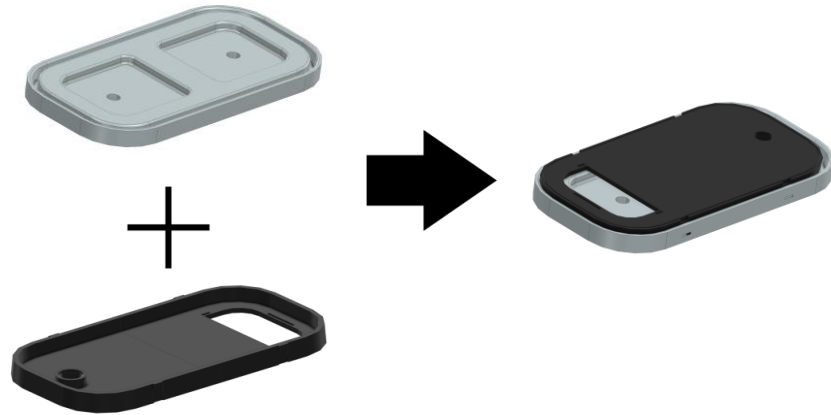
Lid Assembly

1. Ultrasonic Welding



Required Tooling:

- Ultrasonic Welding Fixture



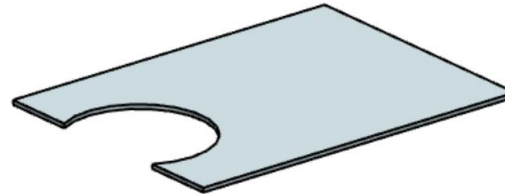
Lid Assembly

2. Sticker Application

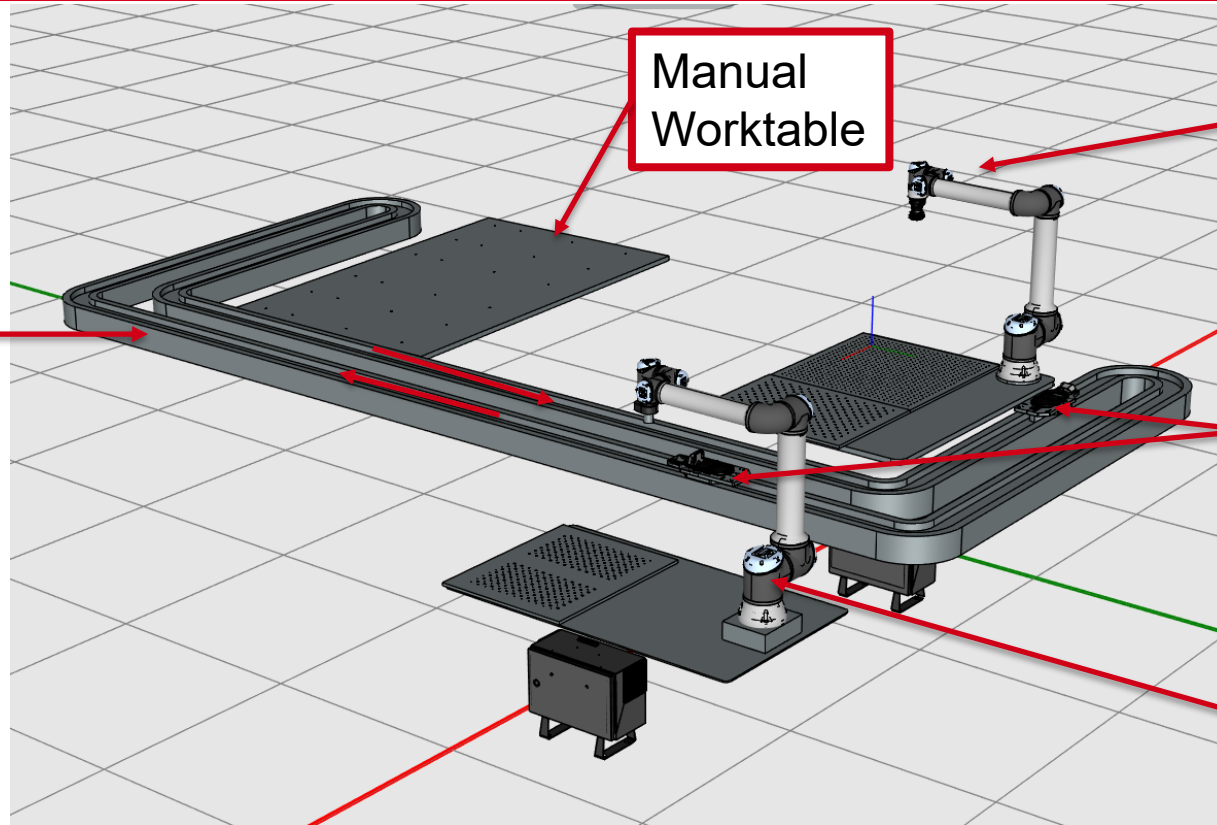


Required Tooling:

- Sticker Application Card



Workplace Layout



Conveyor

Manual
Worktable

Fingered
Robot: Rink
Assembly

Rink Kit at
Conveyor
Station

Vacuum
Robot: Base
Assembly

Goal Assembly

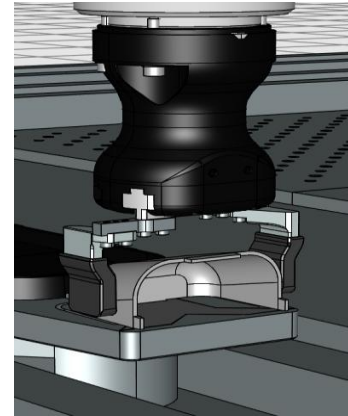
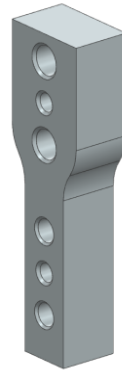
1. Goal + Goalie Press Fit

- Estimated cycle time:
0.75 seconds



Required Tooling:

- Shadowboard Kit
- Gripper finger extenders



Goal Assembly

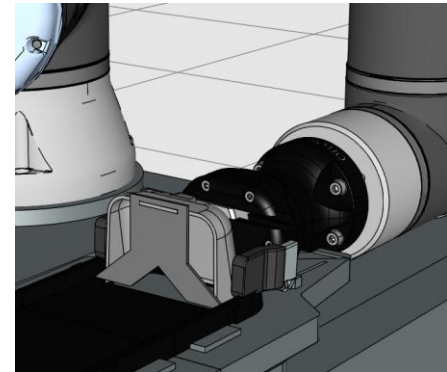
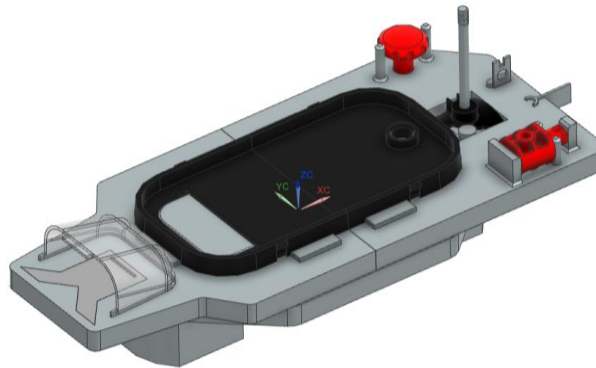
2. Goal to Lid Press Fit

- Estimated cycle time:
8 seconds



Required Tooling:

- Shadowboard Kit
- Gripper finger extenders



Launcher Assembly

1. Spring Attachment

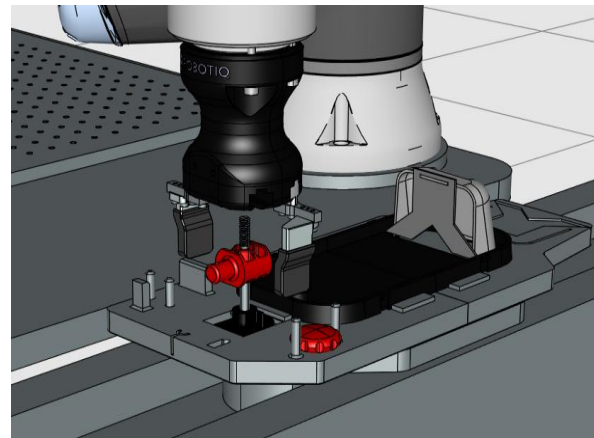
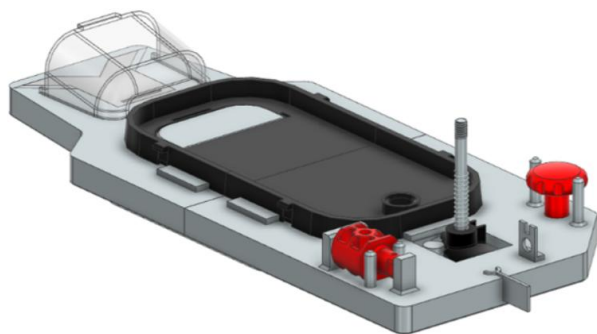


2. Automated Turret Application

- Estimated cycle time: 1.15 seconds

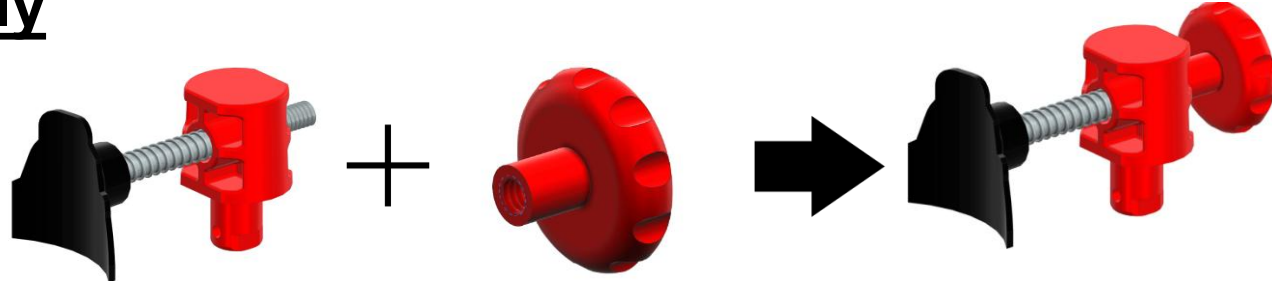
Required Tooling:

- Shadowboard Fixture Kit



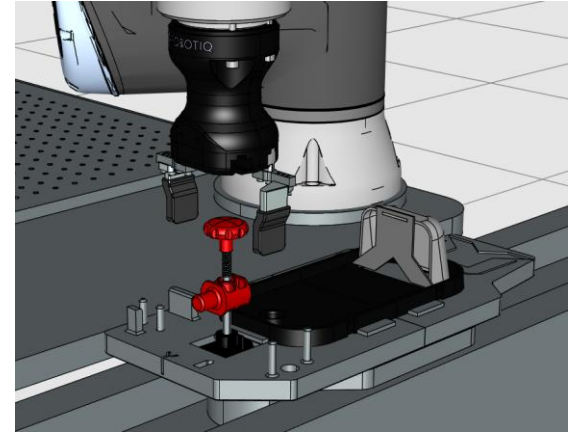
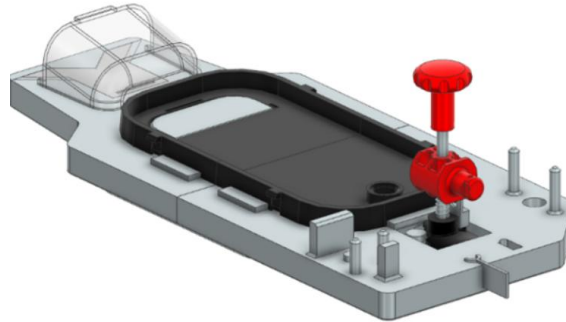
Launcher Assembly

3. Handle Attachment



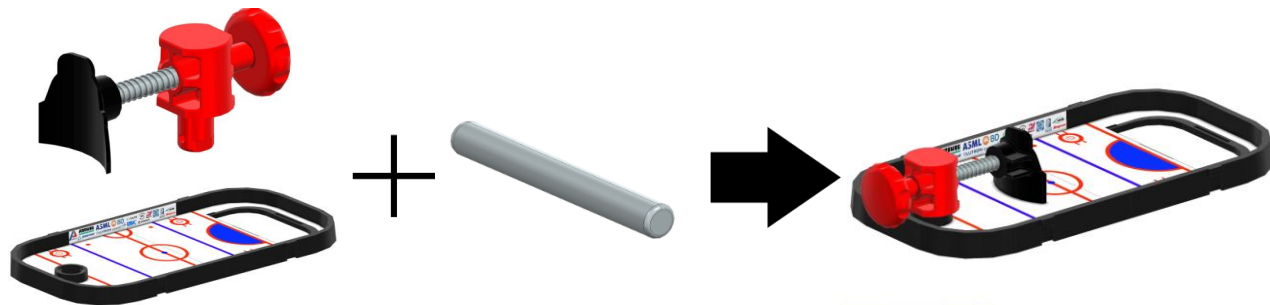
Required Tooling:

- Shadowboard Fixture Kit
- Screw gun end effector



Launcher Assembly

4. Automated Pin Insertion



Required Tooling:

- Launcher Lid Fixture
- Magazine Table Mount
- Pin Magazine
- Pin Hopper



Image 1: Pin Insertion System

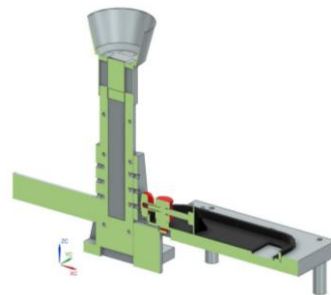


Image 2: Cross Section

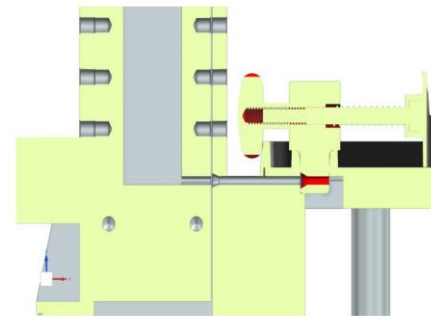
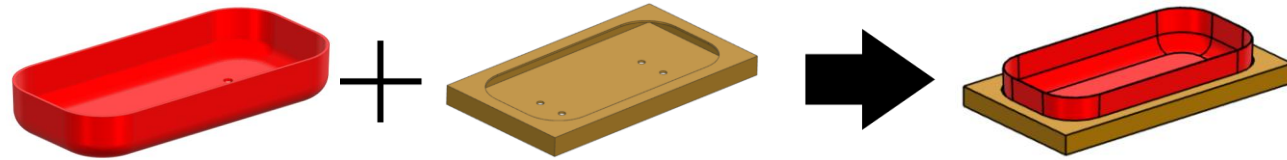


Image 3: Pin Insertion Detailed View

Base Assembly

1. Base + Cardboard Twist Tied



Required Tooling:

- Twist Ties



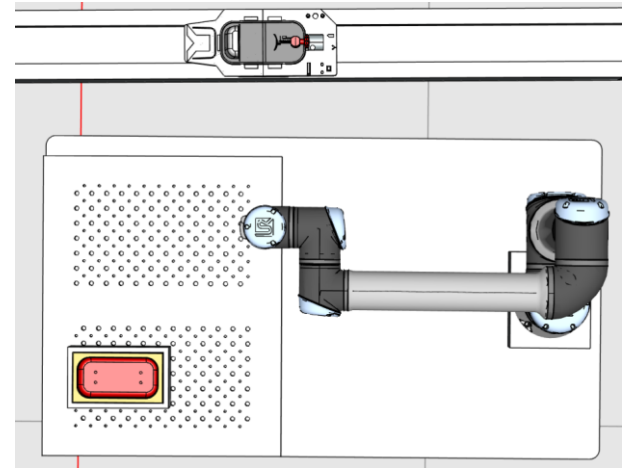
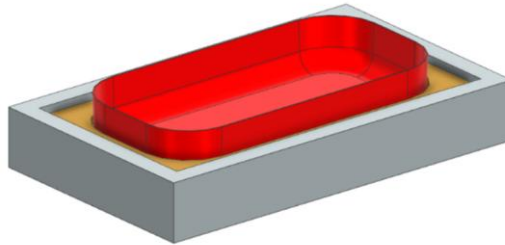
Coin Bank Assembly

1. Base + Lid Press Fit



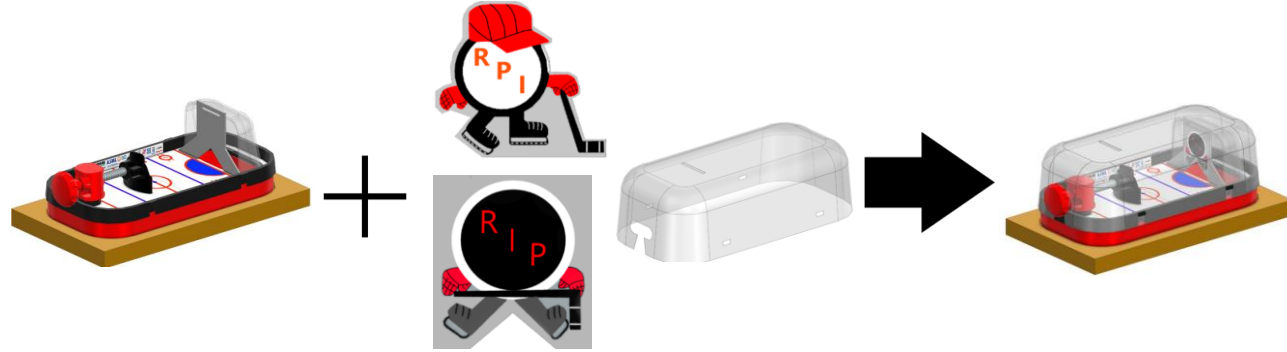
Required Tooling:

- Cardboard base Fixture



Coin Bank Assembly

2. Bubble + Sticker Application



Required Tooling:

- None

Budget Forecast

Part Description	Part #	Material	Supplier	Raw Material Size	Material Size per Unit	Quantity Ordered	Cost Per Item	Total Cost
Launcher Spring	PC1	Music Wire Steel	Century Spring	1.5" free length, 0.3 OD	1	300	\$0.91	\$273.00
Twist Tie	PC3	Paper	Amazon	1800pc	4pc	1	\$9.99	\$9.99
Dowel Pin	PC4	High Strength Steel	McMaster Carr	50pc	1pc	7	\$12.24	\$85.68
Base	C1	A1008 Carbon Steel	Albany Steel	48" X 96" - Thickness: 0.0239	12" x 12"	11	\$39.94	\$439.34
Launcher Handle	C4	Aluminum 6061	Online Metals	40' 1.5" Diameter	1"	3	\$148.54	\$445.62

Budget Forecast

Part Description	Part Number	Material	Supplier	Raw Material Size	Material Size per Unit	Quantity Ordered	Cost Per Item	Total Cost
Goalie	C7	ABS	McMaster Car	24" x 48"	2" x 3"	3	\$27.8	\$83.4
Bubble	C8	PETG	Piedmont Plastic	48" x 96"	16" x 16"	19	\$35	\$665
Stickers	C10, C11, C12, C13	Vinyl	Sign Warehouse	20" x 150' 3-mil	12" x 4"	4	\$110.27	\$441.08
Launcher Bolt	C14	Aluminum 6061	McMaster Carr	12'	2.675"	7	\$28.88	\$202.16
Mold	TBD	Aluminum 6061	MILL	-	-	-	-	\$500
					Unit Cost:	\$10.48	Final Total:	\$3145.27

Donated Materials

Part	Material	Supplier
Lid	ABS	MILL
Launcher Turret	ABS	MILL
Goal	ABS	MILL
Puckman	ABS	MILL
Cardboard Base	Coated Recycled Board	MILL
Base	Powder Coat	LoDolce

Lessons Learned

Technical Lessons	Team & Process Management
I. Optimizing Parts for Certain Manufacturing Processes	I. Cross-Team Integration Prevents Rework
II. Designing Tooling for Manufacturing Processes	II. Communication and Documentation Efficiency Matter
III. Defining Tolerance Requirements and Inspection Procedures	III. Plan Assembly Early
IV. Designing Mechanisms and Tooling for Automation	IV. Weekly Progress Updates = Accountability

Proposed Project Plan for Spring Semester

January:

- Finalize all assembly and tooling designs
- Order all materials
- Begin manufacturing injection molds and deep drawing tooling
- Pin insertion mechanism
- Screw gun end effector

February:

- Manufacture all quality control gauges
- Begin manufacturing bases, handles, and bolts

March:

- Manufacture all fixtures and assembly mechanisms
- Print stickers
- Anodize handles, powder coat bases
- Begin manufacturing injection molded parts

April:

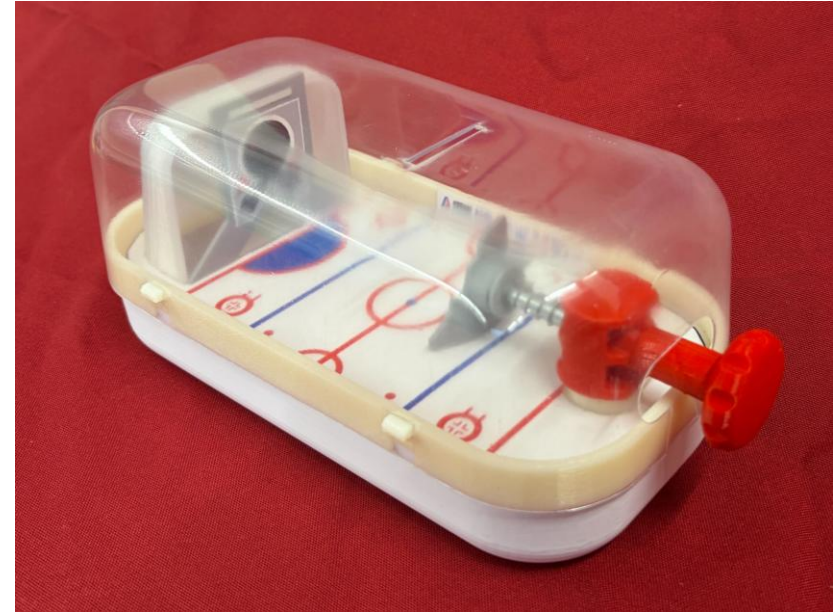
- Manufacture packaging
- Assemble Handheld Hockey

Conclusion

Handheld Hockey will be manufactured using nearly every process in the MILL

Several prototypes have shown the feasibility of the design

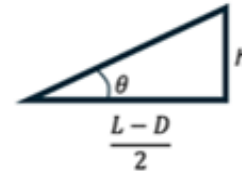
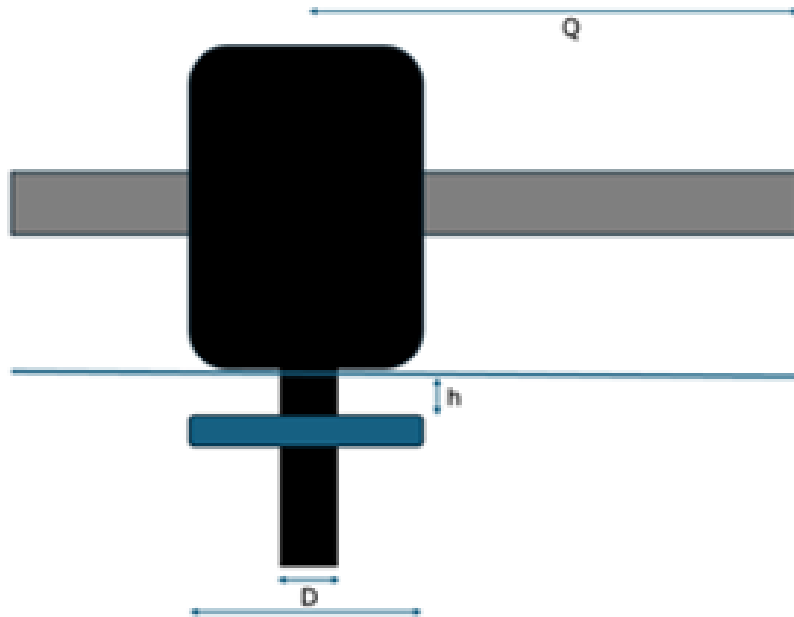
The project meets manufacturing requirements and has remained under budget



Thank You Sponsors!

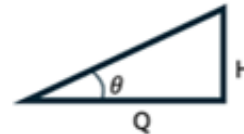


Appendix A: Turret Tolerance Calculations



$$\theta = \arctan \frac{h}{\left(\frac{L-D}{2}\right)} = \arctan \left(\frac{2h}{L-D} \right)$$

$$\tan(\theta) = \frac{2h}{L-D}$$



$$H = Q * \tan(\theta)$$

$$H = \frac{2Qh}{L-D} = \frac{2 * 1.91 * h}{1.0 - 0.5}$$

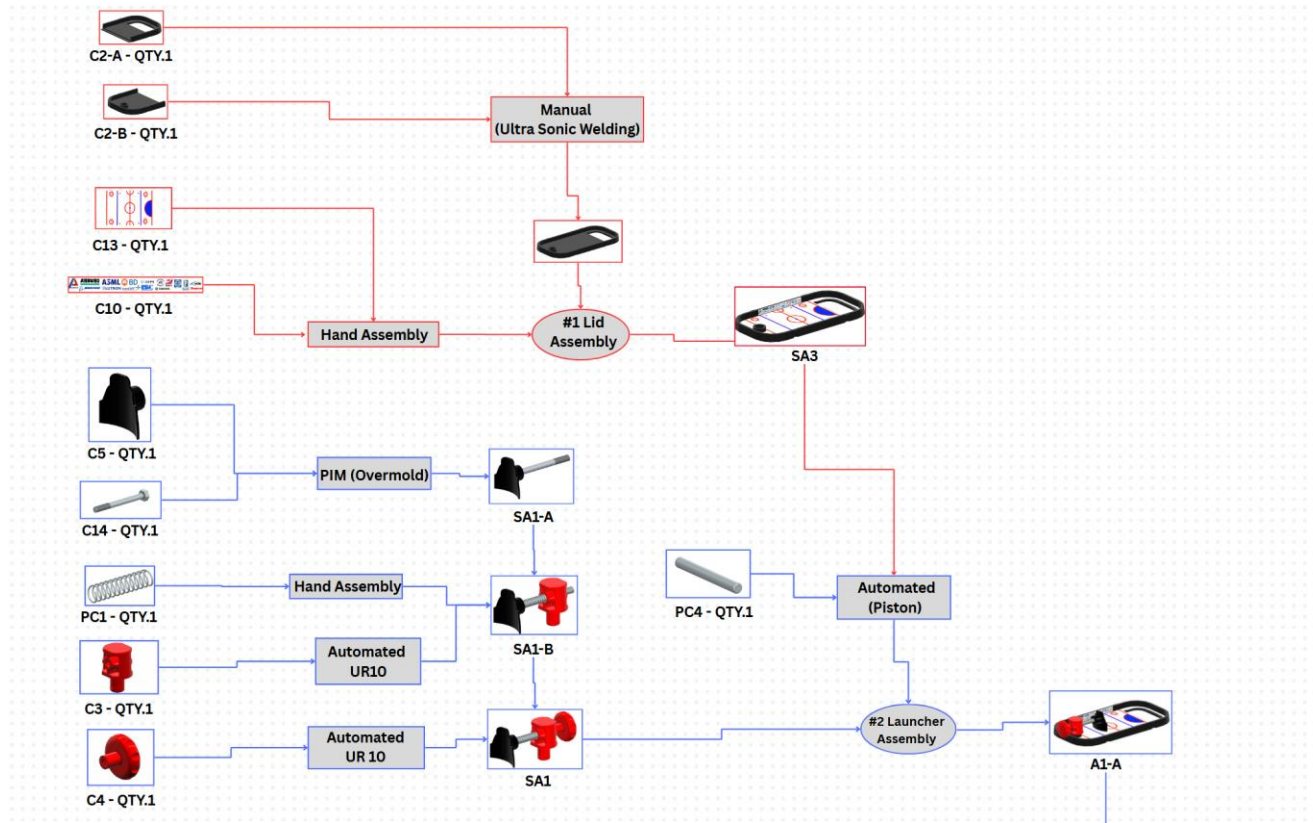
$$H = \frac{2Qh}{L-D} < .027"$$

$$h \leq 0.0035"$$

Appendix B: ANSI RC Fit Clearances

Nominal Size Range, Inches Over To	Class RC1			Class RC2			Class RC3			Class RC4		
	Clear- ance*	Standard Tolerance Limits		Clear- ance*	Standard Tolerance Limits		Clear- ance*	Standard Tolerance Limits		Clear- ance*	Standard Tolerance Limits	
		Hole H5	Shaft g4		Hole H6	Shaft g5		Hole H7	Shaft f6		Hole H8	Shaft f7
		Values shown below are in thousandths of an inch.										
0–0.12	0.1 0.45	+0.2 0	–0.1 –0.25	0.1 0.55	+0.25 0	–0.1 –0.3	0.3 0.95	+0.4 0	–0.3 –0.55	0.3 1.3	+0.6 0	–0.3 –0.7
0.12–0.24	0.15 0.5	+0.2 0	–0.15 –0.3	0.15 0.65	+0.3 0	–0.15 –0.35	0.4 1.12	+0.5 0	–0.4 –0.7	0.4 1.6	+0.7 0	–0.4 –0.9
0.24–0.40	0.2 0.6	+0.25 0	–0.2 –0.35	0.2 0.85	+0.4 0	–0.2 –0.45	0.5 1.5	+0.6 0	–0.5 –0.9	0.5 1.0	+0.9 0	–0.5 –1.1
0.40–0.71	0.25 0.75	+0.3 0	–0.25 –0.45	0.25 0.95	+0.4 0	–0.25 –0.55	0.6 1.7	+0.7 0	–0.6 –1.0	0.6 2.3	+1.0 0	–0.6 –1.3
0.71–1.19	0.3 0.95	+0.4 0	–0.3 –0.55	0.3 1.2	+0.5 0	–0.3 –0.7	0.8 2.1	+0.8 0	–0.8 –1.3	0.8 2.8	+1.2 0	–0.8 –1.6
1.19–1.97	0.4 1.1	+0.4 0	–0.4 –0.7	0.4 1.4	+0.6 0	–0.4 –0.8	1.0 2.6	+1.0 0	–1.0 –1.6	1.0 3.6	+1.6 0	–1.0 –2.0
1.97–3.15	0.4 1.2	+0.5 0	–0.4 –0.7	0.4 1.6	+0.7 0	–0.4 –0.9	1.2 3.1	+1.2 0	–1.2 –1.9	1.2 4.2	+1.8 0	–1.2 –2.4
3.15–4.73	0.5 1.5	+0.6 0	–0.5 –0.9	0.5 2.0	+0.9 0	–0.5 –1.1	1.4 3.7	+1.4 0	–1.4 –2.3	1.4 5.0	+2.2 0	–1.4 –2.8
4.73–7.09	0.6 1.8	+0.7 0	–0.6 –1.1	0.6 2.3	+1.0 0	–0.6 –1.3	1.6 4.2	+1.6 0	–1.6 –2.6	1.6 5.7	+2.5 0	–1.6 –3.2
7.09–9.85	0.6 2.0	+0.8 0	–0.6 –1.2	0.6 2.6	+1.2 0	–0.6 –1.4	2.0 5.0	+1.8 0	–2.0 –3.2	2.0 6.6	+2.8 0	–2.0 –3.8
9.85–12.41	0.8 2.3	+0.9 0	–0.8 –1.4	0.8 2.9	+1.2 0	–0.8 –1.7	2.5 5.7	+2.0 0	–2.5 –3.7	2.5 7.5	+3.0 0	–2.5 –4.5
12.41–15.75	1.0 2.7	+1.0 0	–1.0 –1.7	1.0 3.4	+1.4 0	–1.0 –2.0	3.0 6.6	+2.2 0	–3.0 –4.4	3.0 8.7	+3.5 0	–3.0 –5.2
15.75–19.69	1.2 3.0	+1.0 0	–1.2 –2.0	1.2 3.8	+1.6 0	–1.2 –2.2	4.0 8.1	+2.5 0	–4.0 –5.6	4.0 10.5	+4.0 0	–4.0 –6.5

Appendix - Assembly Process



Appendix - Assembly Process

